Create An Analytical Dataset – Pawdacity Report

Part 3 - Data Cleanup

ABSTRACT

Throughout this report I will demonstrate every taken step to create an analytical dataset for Pawdacity Pet Store. That includes data understanding, data cleansing, data formatting, data blending, and dealing with null/outliers. The output cleaned dataset will be used later for creating a predictive model.

Maged Baheig

Problem Solving with Advanced Analytics-Udacity/Bertelsmann2021

Step 1: Business and Data Understanding

Provide an explanation of the key decisions that need to be made. (250 word limit)

Key Decisions:

Answer these questions

1. What decisions needs to be made?
2. What data is needed to inform those decisions?

Pawdacity is a **leading pet store chain** in **Wyoming** with **13 stores** throughout the state. This year, Pawdacity would like to expand and **open a 14th store**. I'll perform an analysis to **recommend the city** for Pawdacity's **newest store**, based on predicted yearly sales.

The city selection for the **14th store** will be determined based on the **expected yearly sales**, with below considerations:

1. The new store should be located in a city where there are **no current Pawdacity stores**.
2. The new city has a **population over 4,000 people** (based upon the 2014 US Census estimate).
3. The predicted **yearly sales must be over $200,000**.
4. The city chosen has the **highest predicted sales** from the predicted set.
5. The total **sales** for the entire **competition** in the new city should be **less than $500,000**

Key Decisions:

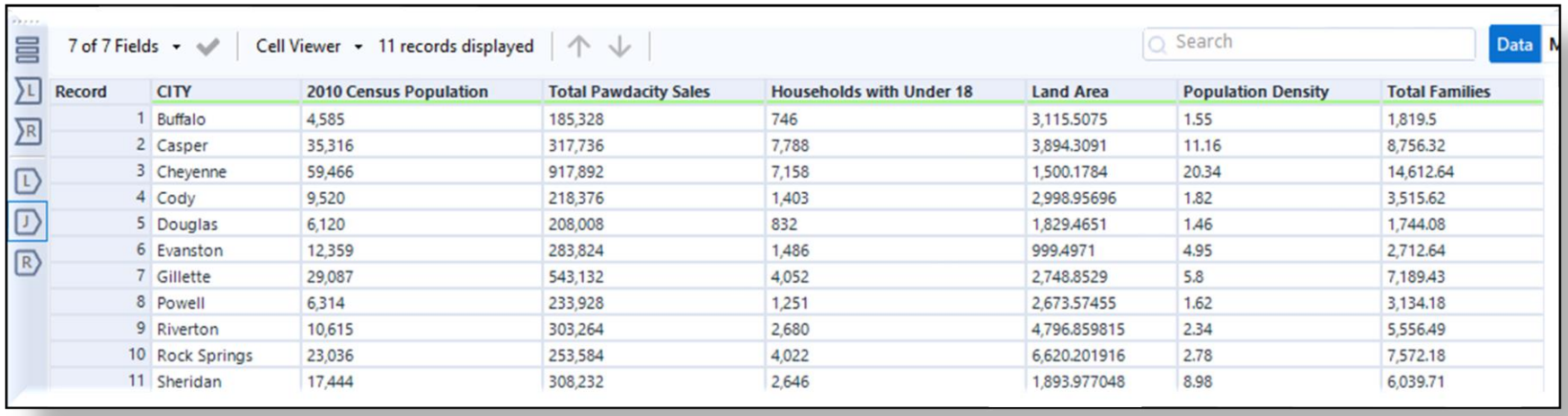
1. **Decision**: What is the **best city** for Pawdacity's **newest 14th store**.
2. **Needed Info**: Predicted Sales for each available City option.
3. **Analysis Type**: **Predicative** analysis to **predict the expected Sales** for each available City option.
4. **Needed Data:** Any variables (**critical factors**) may affect "**Yearly Sales**".

This dataset will be your training set to help you build a regression model in order to predict sales in the Practice Project in the next lesson. Every row should have sales data because we're trying to predict sales.

<u>Step 2</u>: Building the Training Set

Build your training set given the data provided to you. Your column sums of your dataset should match the sums in the table below. ***In addition provide the averages on your data set here to help reviewers check your work. You should round up to two decimal places, ex: 1.24***

Column	Sum	My Solution Sum	Average
Census Population	*213,862*	213,862	19,442
Total Pawdacity Sales	*3,773,304*	3,773,304	343,027.64
Households with Under 18	*34,064*	34,064	3,096.73
Land Area	*33,071*	33,071.38	3,006.49
Population Density	*63*	62.80	5.71
Total Families	*62,653*	62,652.79	5,695.71



Record	CITY	2010 Census Population	Total Pawdacity Sales	Households with Under 18	Land Area	Population Density	Total Families
1	Buffalo	4,585	185,328	746	3,115.5075	1.55	1,819.5
2	Casper	35,316	317,736	7,788	3,894.3091	11.16	8,756.32
3	Cheyenne	59,466	917,892	7,158	1,500.1784	20.34	14,612.64
4	Cody	9,520	218,376	1,403	2,998.95696	1.82	3,515.62
5	Douglas	6,120	208,008	832	1,829.4651	1.46	1,744.08
6	Evanston	12,359	283,824	1,486	999.4971	4.95	2,712.64
7	Gillette	29,087	543,132	4,052	2,748.8529	5.8	7,189.43
8	Powell	6,314	233,928	1,251	2,673.57455	1.62	3,134.18
9	Riverton	10,615	303,264	2,680	4,796.859815	2.34	5,556.49
10	Rock Springs	23,036	253,584	4,022	6,620.201916	2.78	7,572.18
11	Sheridan	17,444	308,232	2,646	1,893.977048	8.98	6,039.71

A Screenshot of ready training dataset

<u>Step 3</u>: Dealing with Outliers

Answer these questions

Are there any cities that are outliers in the training set? Which outlier have you chosen to remove or impute? Because this dataset is a small data set (11 cities), **you should only remove or impute one outlier**. Please explain your reasoning.

- ***<u>As demonstrated at below screenshot</u>***, <u>I've found **2 cities** with **outlier total sales amount**:</u>
  - **Cheyenne** city with **917892$** total sales amounts.
  - **Gillette** city with **543132$** total sales amounts.

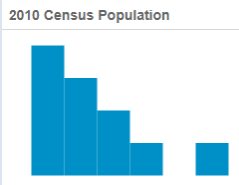
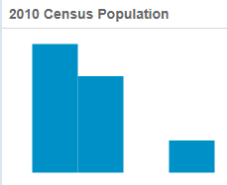
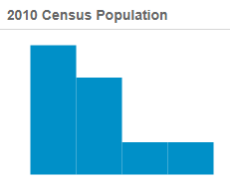
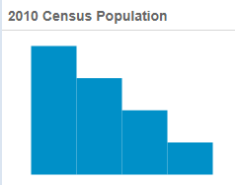
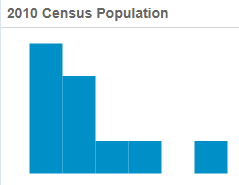
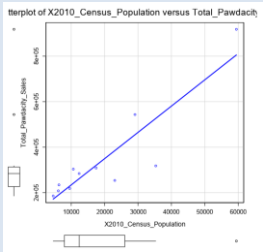
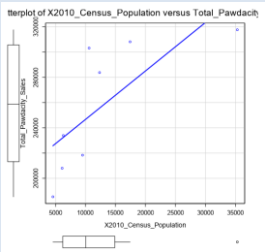
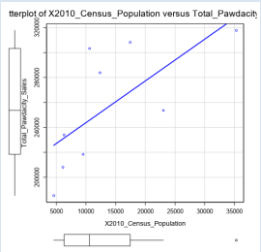
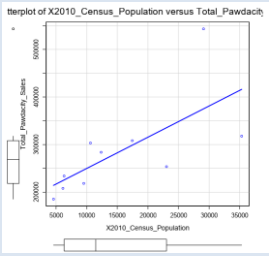
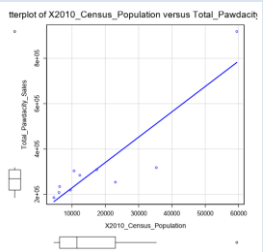
Record	CITY	2010 Census Population	Census_Outlier	Total Pawdacity Sales	Sales_Outlier	Households with Under 18	Under 18_Outlier	Land Area	Area_Outlie
1	Buffalo	4,585	Normal	185,328	Normal	746	Normal	3,115.5075	Normal
2	Casper	35,316	Normal	317,736	Normal	7,788	Normal	3,894.3091	Normal
3	Cheyenne	59,466	Outlier	917,892	Outlier	7,158	Normal	1,500.1784	Normal
4	Cody	9,520	Normal	218,376	Normal	1,403	Normal	2,998.95696	Normal
5	Douglas	6,120	Normal	208,008	Normal	832	Normal	1,829.4651	Normal
6	Evanston	12,359	Normal	283,824	Normal	1,486	Normal	999.4971	Normal
7	Gillette	29,087	Normal	543,132	Outlier	4,052	Normal	2,748.8529	Normal
8	Powell	6,314	Normal	233,928	Normal	1,251	Normal	2,673.57455	Normal
9	Riverton	10,615	Normal	303,264	Normal	2,680	Normal	4,796.859815	Normal
10	Rock Springs	23,036	Normal	253,584	Normal	4,022	Normal	6,620.201916	Outlier
11	Sheridan	17,444	Normal	308,232	Normal	2,646	Normal	1,893.977048	Normal

- <u>My final decision regarding these 2 outliers is</u>:
  - **Keep** Cheyenne city with **917892**$ total sales amounts.
  - **Remove Gillette** city with **543132$** total sales amounts.

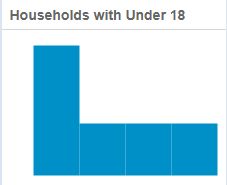
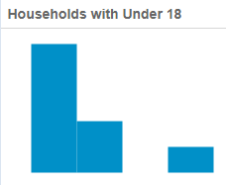
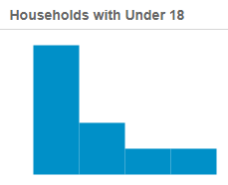
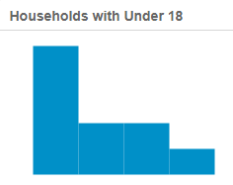
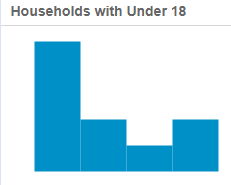
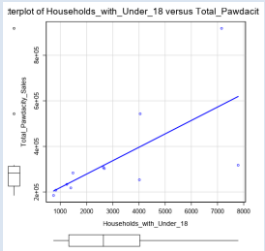
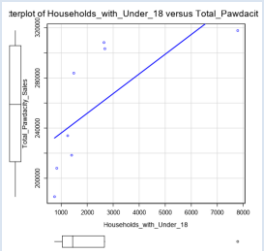
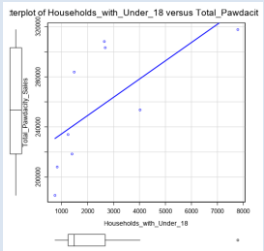
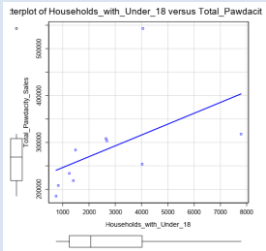
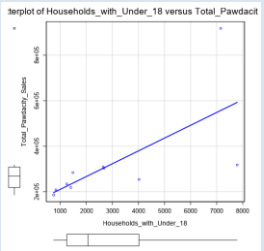
- ***<u>As demonstrated at below summary tables</u>***, <u>I made this decision because that guarantee below for most of our potential predictor variables</u>:
  - **Strong** Relationship with **high** correlation value.
  - **Low** P-Value and **High** R-Squared value.

- ***<u>Note</u>***: *Imputation using Average and Median values (**<u>before</u> & <u>after</u> removing Gillette city**) didn't give better results, that's why I decided not to impute outliers. (**Please check Alteryx workflow "Only includes imputation after removing Gillette city"**)*

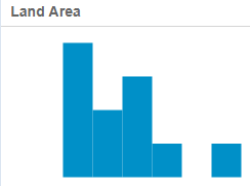
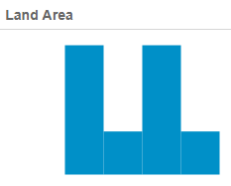
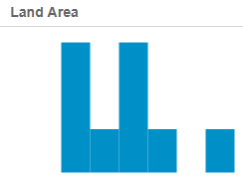
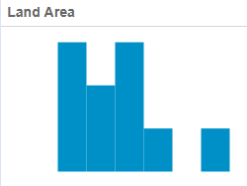
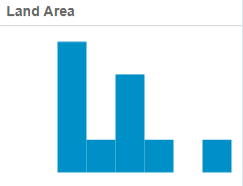
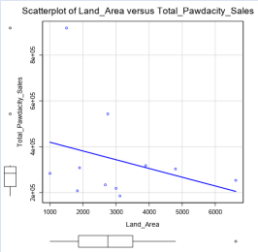
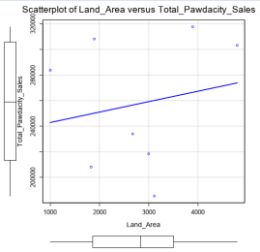
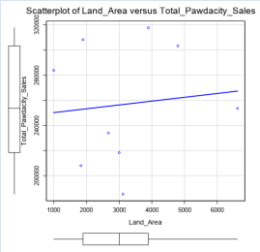
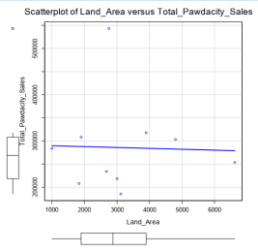
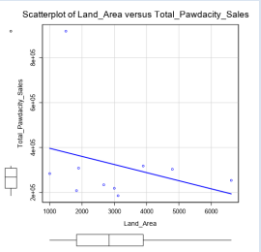
1. 2010 Census Population

Variable Characteristics	Data WITH Outliers [Original Dataset]	Data WithOUT All Fields' Outliers	Data WithOUT Sales Outliers [Cheyenne & Gillette]	Data WithOUT Sales Outliers [Cheyenne]	Data WithOUT Sales Outliers [Gillette]
Distribution Shape					
Distribution Type	***Right*** *Skewed*	***Right*** *Skewed*	***Right*** *Skewed*	***Right*** *Skewed*	***Right*** *Skewed*
Mean	**19442**	**12784.13**	**13923.22**	**15439.6**	**18477.5**
Median	**12359**	**10067.5**	**10615**	**11487**	**11487**
Best Imputation Measure	***Median***	***Median***	***Median***	***Median***	***Median***
Upper Fence	**53278.25**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
Lower Fence	**-19299.75 ~ Zero**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
# of Dataset Records	**11**	**8**	**9**	**10**	**10**
Number of Outliers	**1**	***0***	***0***	***0***	**1**
Outliers Value	**59466**	***N/A***	***N/A***	***N/A***	**59466**
Relation Shape with Total Pawdacity Sales					
Relation	**Very Good Positive**	**Good Positive**	**Low Positive**	**Low Positive**	**Very Good Positive**
Correlation	**0.89810**	**0.74227**	**0.68800**	**0.682255**	**0.89875**
P-Value	**0.00017**	**0.03495**	**0.0405**	**0.02974**	**0.00041**
R-Squared	**0.8066**	**0.551**	**0.4733**	**0.4655**	**0.8078**
Predictor	**Good Predictor**	**Bad Predictor**	**Bad Predictor**	**Bad Predictor**	**Good Predictor**

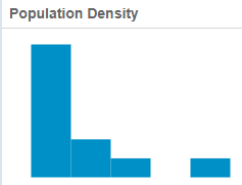
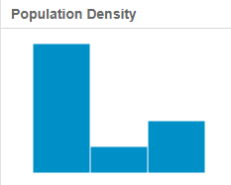
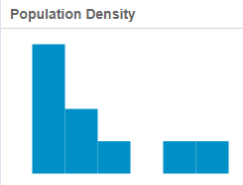
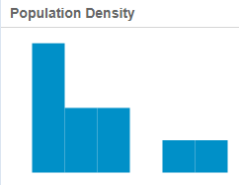
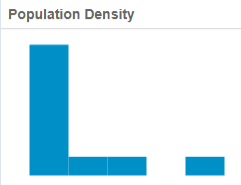
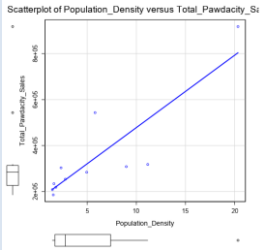
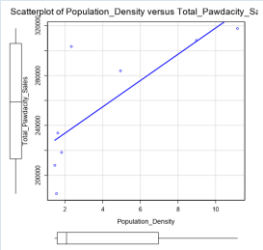
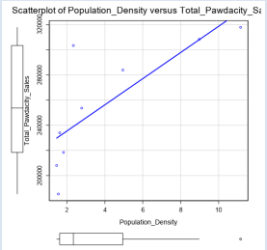
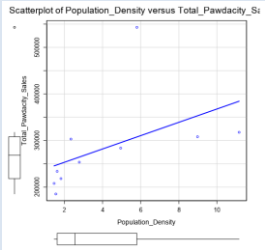
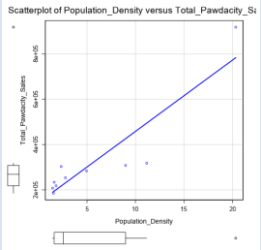
2. Households with Under 18

Variable Characteristics	Data WITH Outliers	Data WithOUT All Fields' Outliers	Data WithOUT Sales Outliers [Cheyenne & Gillette]	Data WithOUT Sales Outliers [Cheyenne]	Data WithOUT Sales Outliers [Gillette]
Distribution Shape					
Distribution Type	***Right** Skewed*	***Right** Skewed*	***Right** Skewed*	***Right** Skewed*	***Right** Skewed*
Mean	**3096.73**	**2354**	**2539.33**	**2690.6**	**3001.2**
Median	**2646**	**1444.5**	**1486**	**2066**	**2066**
Best Imputation Measure	***Median***	***Median***	***Median***	***Median***	***Median***
Upper Fence	**8102**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
Lower Fence	**-2738 ~ Zero**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
# of Dataset Records	**11**	**8**	**9**	**10**	**10**
Number of Outliers	**0**	***0***	***0***	***0***	***0***
Outliers Value	**N/A**	***N/A***	***N/A***	***N/A***	***N/A***
Relation Shape with Total Pawdacity Sales					
Relation	**Low Positive**	**Good Positive**	**Low Positive**	**Very Low Positive**	**Low Positive**
Correlation	**0.67601**	**0.70251**	**0.67378**	**0.493491**	**0.67465**
P-Value	**0.0224**	**0.05201**	**0.04661**	**0.1472**	**0.03236**
R-Squared	**0.457**	**0.4935**	**0.454**	**0.2435**	**0.4552**
Predictor	**Bad Predictor**	**Bad Predictor**	**Bad Predictor**	**Bad Predictor**	**Bad Predictor**

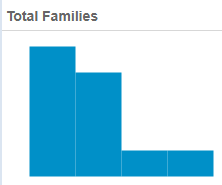
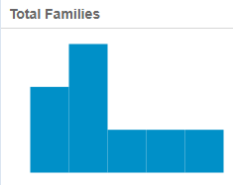
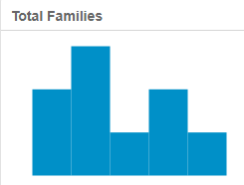
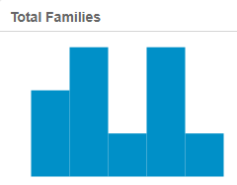
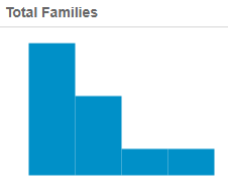
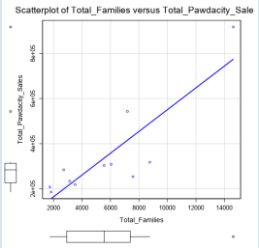
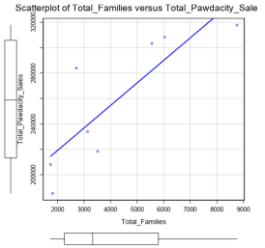
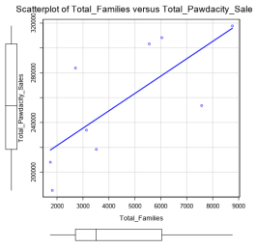
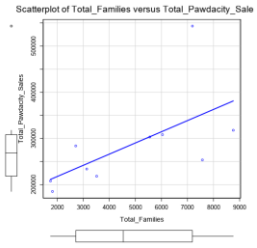
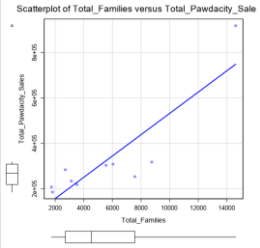
3. Land Area

Variable Characteristics	Data WITH Outliers	Data WithOUT All Fields' Outliers	Data WithOUT Sales Outliers [Cheyenne & Gillette]	Data WithOUT Sales Outliers [Cheyenne]	Data WithOUT Sales Outliers [Gillette]
Distribution Shape					
Distribution Type	***Right** Skewed*	***Right** Skewed*	***Bi-Modal***	***Bi-Modal***	***Right** Skewed*
Mean	**3006.49**	**2775.27**	**3202.48**	**3157.12**	**3032.25**
Median	**2748.85**	**2836.27**	**2998.96**	**2873.90**	**2836.27**
Best Imputation Measure	***Median***	***Median***	***Average***	***Average***	***Median***
Upper Fence	**5969.69**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
Lower Fence	**-603.06 ~ Zero**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
# of Dataset Records	**11**	**8**	**9**	**10**	**10**
Number of Outliers	**1**	***0***	**1**	**1**	**1**
Outliers Value	**6620.20**	***N/A***	**6620.20**	**6620.20**	**6620.20**
Relation Shape with Total Pawdacity Sales					
Relation	**Very Weak Negative**	**Very Weak Positive**	**Very Weak Positive**	**No Relationship**	**Very Weak Negative**
Correlation	**-0.28890**	**0.19169**	**0.10783**	**-0.030629**	**-0.28708**
P-Value	**0.3889**	**0.64929**	**0.78245**	**0.93306**	**0.42126**
R-Squared	**0.08346**	**0.03675**	**0.01163**	**0.000938**	**0.08241**
Predictor	**Bad Predictor**	**Bad Predictor**	**Bad Predictor**	**Bad Predictor**	**Bad Predictor**

4. Population Density

Variable Characteristics	Data WITH Outliers	Data WithOUT All Fields' Outliers	Data WithOUT Sales Outliers [Cheyenne & Gillette]	Data WithOUT Sales Outliers [Cheyenne]	Data WithOUT Sales Outliers [Gillette]
Distribution Shape					
Distribution Type	***Right*** *Skewed*	***Right*** *Skewed*	***Right*** *Skewed*	***Right*** *Skewed*	***Right*** *Skewed*
Mean	**5.7**	**4.24**	**4.07**	**4.24**	**5.7**
Median	**2.8**	**2.08**	**2.34**	**2.56**	**2.56**
Best Imputation Measure	***Median***	***Median***	***Median***	***Median***	***Median***
Upper Fence	**15.9**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
Lower Fence	**-6.8 ~ Zero**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
# of Dataset Records	**11**	**8**	**9**	**10**	**10**
Number of Outliers	**1**	***0***	***0***	***0***	**1**
Outliers Value	**20.34**	***N/A***	***N/A***	***N/A***	**20.34**
Relation Shape with Total Pawdacity Sales					
Relation	**Good Positive**	**Good Positive**	**Good Positive**	**Weak Positive**	**Very Good Positive**
Correlation	**0.86289**	**0.77868**	**0.77482**	**0.486034**	**0.90618**
P-Value	**0.00063**	**0.0228**	**0.0142**	**0.15436**	**3e-04**
R-Squared	**0.7446**	**0.6063**	**0.6004**	**0.2362**	**0.8212**
Predictor	**Good Predictor**	**Good Predictor**	**Good Predictor**	**Bad Predictor**	**Good Predictor**

5. Total Families

Variable Characteristics	Data WITH Outliers	Data WithOUT All Fields' Outliers	Data WithOUT Sales Outliers [Cheyenne & Gillette]	Data WithOUT Sales Outliers [Cheyenne]	Data WithOUT Sales Outliers [Gillette]
Distribution Shape					
Distribution Type	***Right*** *Skewed*	***Right*** *Skewed*	***Right*** *Skewed*	***Bi-Modal***	***Right*** *Skewed*
Mean	**5695.70**	**4159.82**	**4538.97**	**4804.02**	**5546.33**
Median	**5556.49**	**3324.9**	**3515.62**	**4536.06**	**4536.06**
Best Imputation Measure	***Median***	***Median***	***Median***	***Average***	***Median***
Upper Fence	**14066.90**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
Lower Fence	**-3762.68 ~ Zero**	*No New Calculation*	*No New Calculation*	*No New Calculation*	*No New Calculation*
# of Dataset Records	**11**	**8**	**9**	**10**	**10**
Number of Outliers	**1**	***0***	***0***	***0***	**1**
Outliers Value	**14612.64**	***N/A***	***N/A***	***N/A***	**14612.64**
Relation Shape with Total Pawdacity Sales					
Relation	**Very Good Positive**	**Very Good Positive**	**Good Positive**	**Low Positive**	**Very Good Positive**
Correlation	**0.86466**	**0.83476**	**0.73498**	**0.606820**	**0.87466**
P-Value	**0.00059**	**0.00993**	**0.02407**	**0.06285**	**0.00093**
R-Squared	**0.7476**	**0.6968**	**0.5402**	**0.3682**	**0.765**
Predictor	**Good Predictor**	**Good Predictor**	**Bad Predictor**	**Bad Predictor**	**Good Predictor**

Before you Submit

Please check your answers against the requirements of the project dictated by the rubric here. Reviewers will use this rubric to grade your project.

Thank You